

Supplementary Methods, Diagnostics and Replication Inventory

Supplementary Information for the anonymous manuscript “Regulatory Resilience in Technology Transfer Systems”.

S1 Corpus and preprocessing

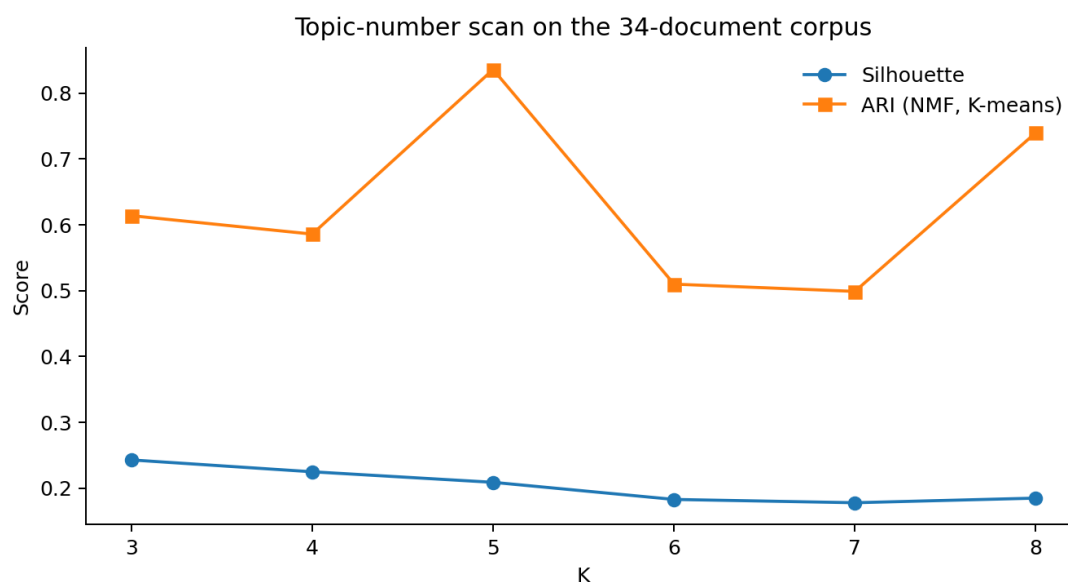
The corpus contains 34 Vietnamese-language files: 27 news or policy documents and 7 legal or professional documents. Each file is treated as one document. The inventory supplied with the replication package provides the anonymous document identifier, filename, operational genre flag and dual-role-source flag. Full texts are withheld from the public package because they reproduce third-party publications.

Preprocessing lower-cases text, removes URLs, expands selected unambiguous abbreviations, uses lexicon-based maximum-matching tokenisation with domain multi-word terms, and removes general and structural stop words. TF-IDF uses $\text{min_df}=2$ and $\text{max_df}=0.9$. NMF uses `nndsvda` initialization, seed 42 and 500 maximum iterations. K-means uses the same TF-IDF matrix, seed 42 and ten initializations.

S2 K scan

K	Cosine silhouette	Largest topic (%)	ARI NMF–K-means
3	0.243	55.9	0.614
4	0.225	52.9	0.586
5	0.209	44.1	0.836
6	0.183	38.2	0.510
7 retained	0.178	38.2	0.499
8	0.185	38.2	0.740

Table S1 Topic-number scan on the 34-document corpus



S3 Robustness checks

Leave-one-out: after removing Bùi Thị Hằng Nga (2019), a contract-and-dispute topic remains. The other two contract-law documents join that topic with one news item on foreign-investment transfer practice. The distinctive essential-facilities keywords do not recur.

News/policy only: re-estimation on 27 news and policy documents produces themes for general transfer activity, industrial AI and data, agriculture, semiconductors, university innovation, strategic technology or cybersecurity, and mechanical engineering. No contract-law or essential-facilities theme is identified from the reported keywords.

S4 Replication files

File	Purpose
corpus_inventory.csv	Anonymous inventory and genre flags
scripts/run_robustness.py	Documented TF-IDF, NMF, K scan, robustness checks and MDS
results/doc_assignment_k7.csv	Full-corpus $K=7$ assignments
results/doc assignment leave one out bui2019.csv	Leave-one-out assignments
results/topics_k7.csv	Full-corpus topic keywords
results/topics leave one out bui2019.csv	Leave-one-out topic keywords
results/topics_news_policy_only.csv	News/policy-only topic keywords
results/topics legal scholarly only.csv	Legal/professional-only topic keywords
results/k_scan.csv	$K=3-8$ diagnostics
results/fig_mds.png	MDS figure used in the manuscript

S5 Reproducibility boundary

The public outputs can be inspected without the copyrighted full texts. Exact re-estimation requires access to those texts and the preprocessing lexicons. The package documents the NMF diagnostics and robustness checks; it does not reproduce the earlier LDA model or establish theoretical saturation or inter-coder reliability.